DA15 Exploring relationships: scatter plots, correlation and cause

Week 5, Explore with NumPy and seaborn | Session 15 of 24 | Python for Data Analysis

Today's question. Do busier days bring in more money, and does rain keep customers away?

Objective. By the end: you can draw a scatter plot coloured by group, read a correlation number, and say why a correlation does not prove one thing causes another.

> **WHERE THIS IS USED**
> Taxi parks, shops and farms all ask whether rain changes demand. Two things moving together do not prove that one causes the other. Do rainy days keep customers away, or is something else going on?

Before this session: DA14 Exploring distributions with seaborn. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
correlation	A number from -1 to 1 for how closely two columns move together.	Taller pupils usually have longer feet.
strength	How close the correlation is to 1 or -1. Near 0 means weak.	How tightly the dots hug a line.
heatmap	A table of numbers coloured by size.	A map shaded by rainfall.
hue	A colour that shows which group each dot belongs to.	A different coloured pen for each shop.
cause	One thing makes another happen. A correlation alone does not show it.	Rain makes the ground wet. Wet ground does not make rain.

Block A. A scatter plot and one correlation

sns.scatterplot puts one column on each axis. hue= colours the dots by group. .corr() calculates the correlation between two columns.

Predict. Predict whether the correlation between customers and revenue is close to 0, close to 1 or close to -1. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
daily = pd.read_csv("sunrise_daily_summary.csv")
sns.scatterplot(data=daily, x="customers_counted", y="revenue_ugx", hue="branch")
plt.show()
r = daily["customers_counted"].corr(daily["revenue_ugx"])
print(round(r, 3))
```

Run it, then compare. The dots rise from left to right, so more customers usually mean more revenue. The correlation is 0.623: positive and moderately strong. The dots are scattered, so a customer count does not fix the revenue.

Investigate.

1. What would a correlation of 1 look like on the chart?
2. Why do the colours help?

Modify. Find the correlation between customers and the number of sale lines (sale_lines).

Block B. A heatmap of every pair, and the rain question

A heatmap of .corr() shows every pair at once. annot=True writes the numbers in the squares. Rain is in the data as rain_mm. Does it matter?

Predict. Predict whether rain_mm is strongly related to customers. Write your guess on paper first.

Your guess, written before you run it

```
cols = ["customers_counted", "rain_mm", "sale_lines", "revenue_ugx"]
table = daily[cols].corr()
sns.heatmap(table, annot=True, vmin=-1, vmax=1)
plt.show()
print(round(table.loc["rain_mm", "customers_counted"], 3))
print(round(table.loc["sale_lines", "revenue_ugx"], 3))
```

Run it, then compare. Rain and customers have a correlation of -0.013, which is essentially zero. Sale lines and revenue have 0.565. In this data the weather barely matters. A weak correlation is a finding too.

Investigate.

1. What does -0.013 tell us about rain?
2. Could a strong correlation still be a coincidence?

Modify. Find the correlation of rain_mm with revenue_ugx.

Make

Check the rain question for each shop separately. For each branch find the correlation between rain_mm and customers_counted. Say whether rain matters at any shop and why a correlation could not prove that it does.

Save your work as DA15. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
KeyError: "hue"	The hue column name is not in the table.	Print list(daily.columns) and use the exact name.
The heatmap shows all one colour	vmin and vmax were not set, or only one column was used.	Pass the four numeric columns and set vmin=-1 and vmax=1.

Exit ticket

A correlation tells me how closely two columns move together, and it does not tell me why.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.